MILITARY STANDARD

LOADING OF HAZARDOUS MATERIALS AND ASSOCIATED ITEMS IN MILVAN CONTAINERS

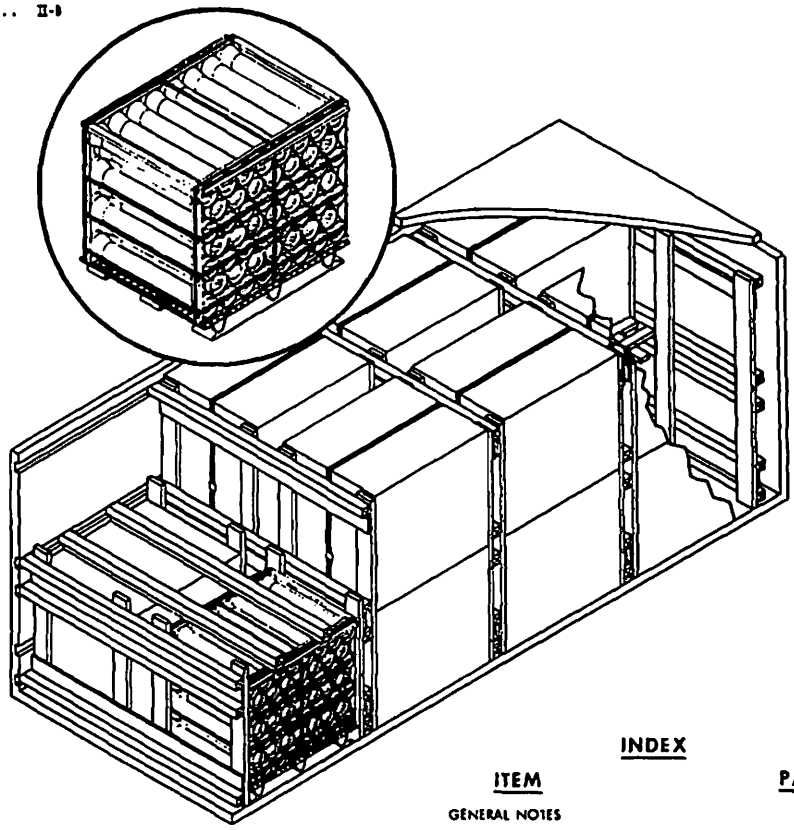
3"/50 CARTRIDGE
FLEET ISSUE UNIT LOAD

MIL-STD-1386-27
(NAVY)

17 SEPTEMBER 1973

UNIT LOAD DATA

UNIT LOAD DRAWING	WR-54/1
WEIGHT	1826 POUNDS
DIMENSIONS	48 7/16 L x 40 W x 42 1/2 H
CUBE	47.6 CU FT
DOT HAZARD CLASSIFICATION	EXPLOSIVES B
COAST GUARD CLASS	II-B



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NOTES:

1. UNLESS OTHERWISE SPECIFIED DIMENSIONS ARE IN INCHES.
2. FOR CROSS REFERENCE TO ASSOCIATED PALLETIZING, TRUCKLOADING AND CARLOADING MILITARY STANDARDS. REFER TO THE INDEX TO STANDARDS, MIL-HDBK-236 (NAVY).

FSC 8140

AUTHORIZED FOR TRAILER-ON-FLAT-CAR (TOFC), CONTAINER-ON-FLAT-CAR (COFC), CONTAINERSHIP AND HIGHWAY SHIPMENTS

M.S. Gray NWHL 9/5/73
SIGNATURE TECHNICAL DIRECTION AGENT (TDA) DATE

J C Kelly
SIGNATURE ORDTYSCOM BY DIRECTION DATE

APPROVED BY US COAST GUARD

DATE

APPROVED BY BUREAU OF EXPLOSIVES

DATE 9/17/73

ORIGINATOR
SIGNATURE 9/5/73

NAVAL WEAPONS HANDLING LABORATORY
N A D EARLE, NEW JERSEY

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GENERAL NOTES

1. THIS DOCUMENT GIVES DETAILED INSTRUCTIONS FOR LOADING THE MILVAN CONTAINER WITH FLEET ISSUE UNIT LOADS OF 3"/50 CARTRIDGES IN MK 5 MOD 0 CARTRIDGE TANKS. THE UNIT LOAD SHALL BE IN ACCORDANCE WITH WR-54/1.
2. THE MILVAN SHALL BE EQUIPPED WITH A MECHANICAL LOAD-BRACING SYSTEM THAT MEETS THE REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6 C.
3. IN ADDITION TO THE BELT RAILS REQUIRED BY PAMPHLET 6C, THE MILVAN CONTAINER HAS BELT RAILS AT THE 72 AND 83 INCH HEIGHTS FROM THE CONTAINER FLOOR.

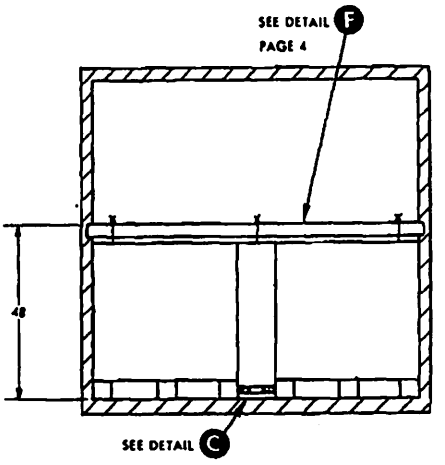
CAUTION

WHEN POSITIONING CROSSMEMBERS MAKE A VISUAL CHECK TO ENSURE THAT CROSSMEMBERS ARE LOCKED SECURELY IN PLACE.

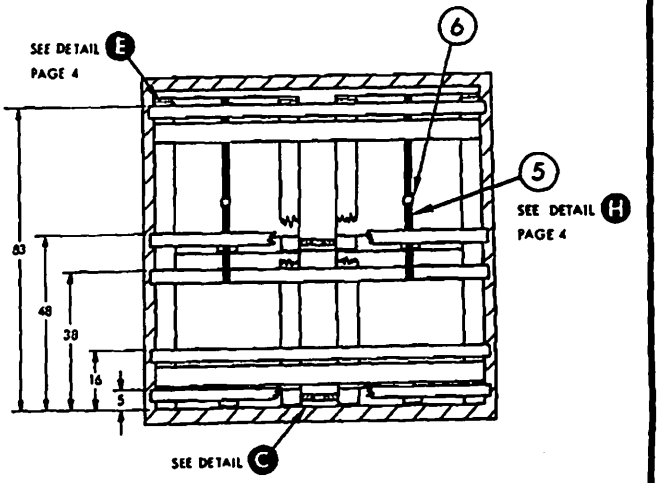
4. CONTAINERS ARE EQUIPPED WITH 25 CROSSMEMBERS. UNUSED CROSSMEMBERS MUST REMAIN WITH THE CONTAINER AND SHALL BE FASTENED INTO THE BELT RAILS FOR SHIPMENT.
5. WHEN THE VOID BETWEEN THE CROSSMEMBER AND THE LOAD IS GREATER THAN ONE INCH, USE FILL MATERIAL TO SHIM BETWEEN CROSSMEMBER AND LOAD AS SHOWN IN DETAIL G, PAGE 4.
6. SHIM, PIECE 4 (DETAIL F), SHALL BE OF SUFFICIENT THICKNESS TO FILL THE VOID BETWEEN THE LADING AND THE BOTTOM OF THE CROSSMEMBER. ONE HALF INCH SLACK IS PERMITTED.
7. SWAY BRACE, DETAIL C, IS ASSEMBLED IN PLACE. PIECE 1 IS POSITIONED INBOARD OF THE OUTER LEGS OF THE PALLET SKIDS. WITH PIECES 2 AGAINST PALLET SKIDS NAIL TO PIECE 1 WITH THREE 10d NAILS PER JOINT. CUT PIECE 3 TO FIT BETWEEN PIECES 2 AND NAIL TO PIECE 1 WITH TWO 10d NAILS.
8. APPLICABLE MATERIAL SPECIFICATIONS:
   DUNNAGE LUMBER - FED. SPEC. MM-L-751
   NAILS - FED. SPEC. FF-N-105, TYPE II, STYLE 10, COMMON BRIGHT
   STRAPPING - FED. SPEC. QQ-S-781, TYPE I, CLASS A.
9. AFTER BLOCKING AND BRACING HAS BEEN INSPECTED, ATTACH SHIPPING DOCUMENTS INSIDE THE CONTAINER IN AN ACCESSIBLE AREA AND CLOSE AND SEAL CONTAINER DOORS AND ATTACH APPLICABLE PLACARDS TO THE OUTSIDE OF THE CONTAINER FOR THE APPROPRIATE MODE OF TRANSPORTATION AS PRESCRIBED IN OP 2165 (VOL. 1).

TOFC/COFC, CONTAINERSHIP AND HIGHWAY

1. THE LOADED MILVAN CONTAINER SHALL BE ON A 20-FT MILVAN CHASSIS EQUIPPED WITH DOUBLE BOGIES WHEN BEING MOVED IN TOFC SERVICE.
2. WHEN THE MILVAN CONTAINER IS SHIPPED TOFC OR COFC, THE RAIL CAR SHALL COMPLY WITH THE REQUIREMENTS OF THE BUREAU OF EXPLOSIVES PAMPHLET 6 C.
3. THE LOADED CONTAINER MAY BE TRANSPORTED OVER PUBLIC HIGHWAYS ON THE MILVAN CHASSIS USING THE FOLLOWING CONFIGURATION:
   A. ONE CONTAINER ON 20-FT MILVAN CHASSIS WITH A DOUBLE BOGIE AND PULLED BY A DOUBLE DRIVE AXLE TRAILER.
4. THE LOADED CONTAINER MAY ALSO BE TRANSPORTED OVER PUBLIC HIGHWAYS USING SUITABLE COMMERCIAL EQUIPMENT OF ADEQUATE CAPACITY AND PULLED BY A DOUBLE DRIVE AXLE TRACTOR. CONTAINERS MUST BE ADEQUATELY SECURED TO THE EQUIPMENT.

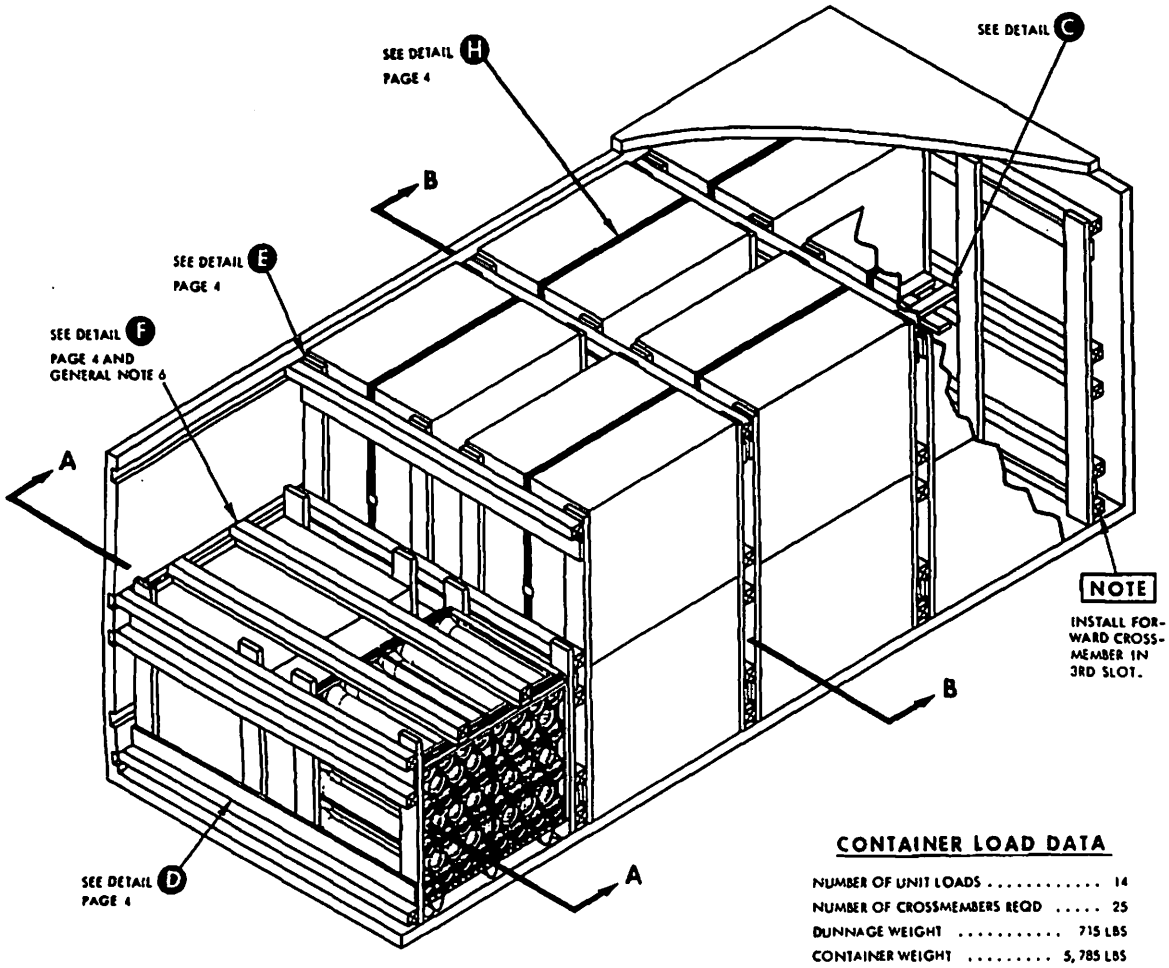


SECTION A-A



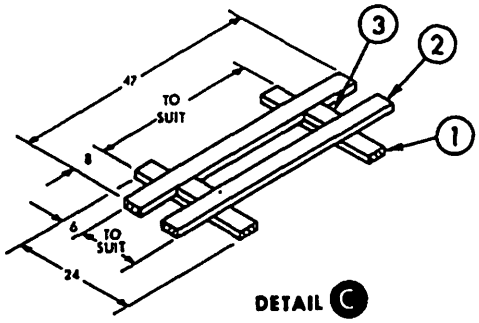
SECTION B-B

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CONTAINER LOAD DATA

NUMBER OF UNIT LOADS	14
NUMBER OF CROSSMEMBERS REQD	25
DUNNAGE WEIGHT	715 LBS
CONTAINER WEIGHT	5,785 LBS
LOAD WEIGHT	25,564 LBS
CONTAINER GROSS WEIGHT	32,064 LBS



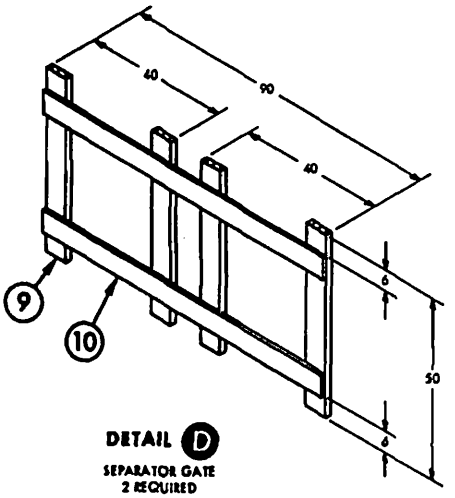
DETAIL C
SWAY BRACE
7 REQUIRED
SEE GENERAL NOTE 7

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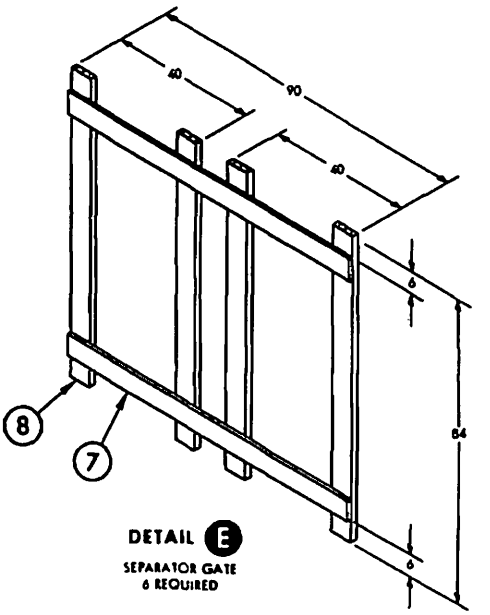
PIECE NO.	DESCRIPTION	SIZE	NO. PCS REQD	NAIL TO	NAILS NUMBER	NAILS SIZE
10	CROSSMEMBER	1 x 6 x 90	4	9	3 PER JOINT	10d[M]
9	VERTICAL MEMBER	2 x 6 x 50	8	SEE 10	-	-
8	VERTICAL MEMBER	2 x 6 x 84	24	SEE 7	-	-
7	CROSSMEMBER	1 x 6 x 90	12	8	3 PER JOINT	10d[M]
6	SEAL	FOR 1 1/4 STRAP	12	-	-	-
5	STRAP	1 1/4 x .035 x 20 FT	6	-	-	-
4	SHIM	4 WIDE x 90 THICKNESS TO SUIT	2	-	-	-
3	SPACER	2 x 4 x TO SUIT	14	1	2	10d
2	STRINGER	2 x 4 x 47	14	1	3 PER JOINT	10d
1	SUPPORT	2 x 4 x 24	14	SEE 2	-	-

LIST OF MATERIALS AND NAILING DATA

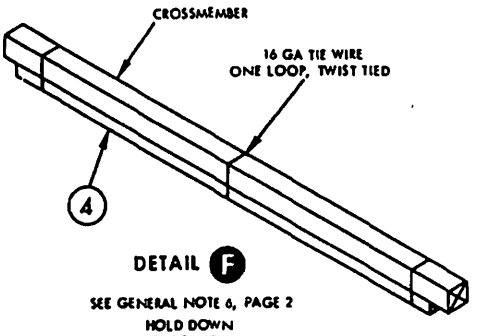
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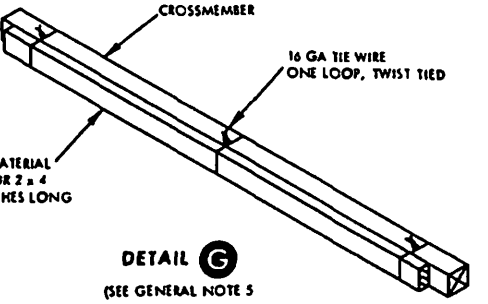
DETAIL D
SEPARATOR GATE
2 REQUIRED



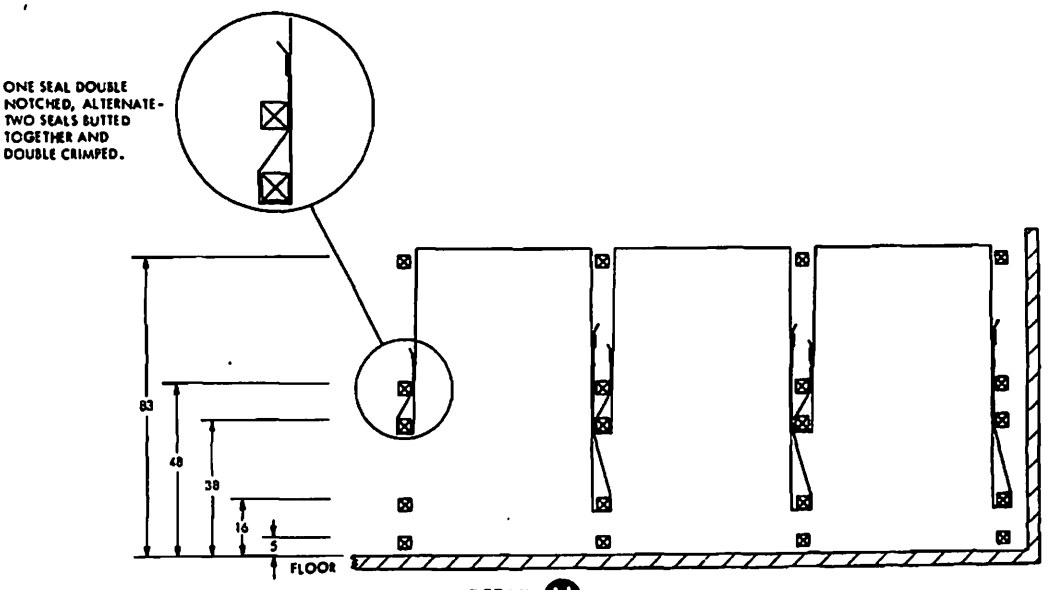
DETAIL E
SEPARATOR GATE
6 REQUIRED



DETAIL F
SEE GENERAL NOTE 6, PAGE 2
HOLD DOWN
2 REQUIRED



DETAIL G
(SEE GENERAL NOTE 5
PAGE 2)



DETAIL H
STRAPPING DIAGRAM
(INDICATES ATTACHMENT POINTS FOR
HOLD DOWN STRAPPING)

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REVIEW ACTIVITY:
NAVY - OS

PREPARING ACTIVITY:
NAVY - OS
(PROJECT NO. 8140-N202)

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